

A general framework for using two-stage IPD-MA to predict individualized treatment effects

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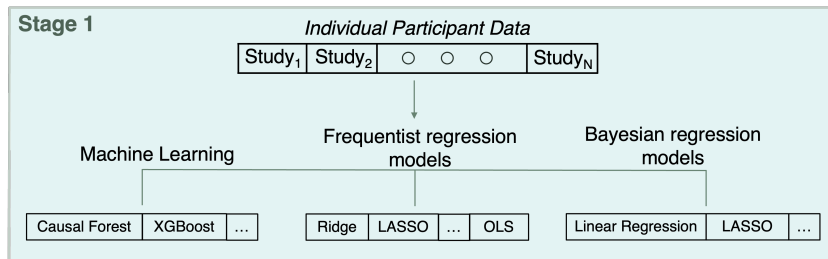
Background and aims

- A key application of IPD-MA is to estimate individualized treatment effect (ITE)
- Even if one stage approach is widely used and conceptually easy, a two stage approach is sometimes necessary

Aims

- ① present a **general** two-stage IPD meta-analysis **framework** for predicting **ITE** for a continuous health outcome
 - **Within trial flexibility** – supports any type of model employable to predict ITE
 - **Between trial adaptability** – different predictive models can be used across trials in the first stage
- ② assess performance of different specifications of the framework with a **simulation study** across multiple scenarios

Framework pipeline: a two-stage approach



First stage: fit study-specific models that can either

- estimate regression coefficients (main treatment effect and effect modification)
- predict ITE

In this stage we considered a complementary range of models:

- Frequentist
- Bayesian
- Machine learning

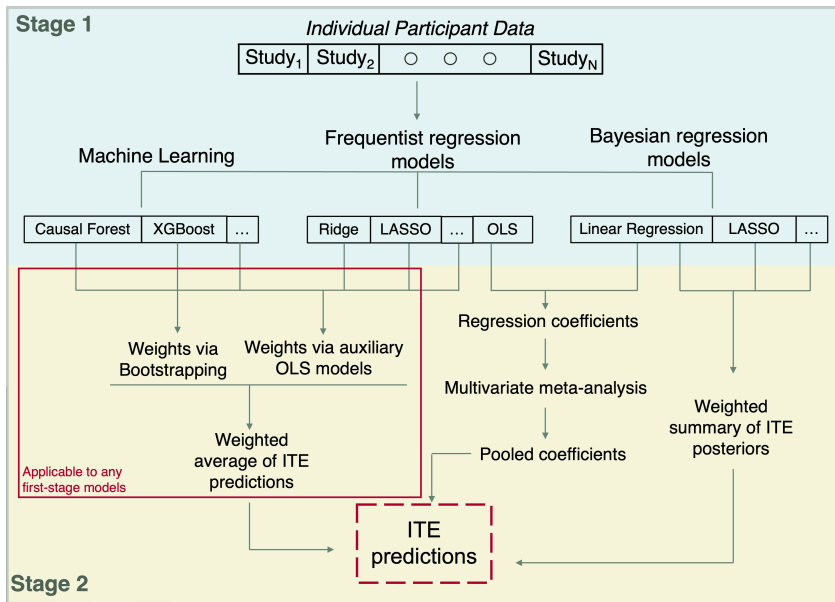
Framework pipeline: a two-stage approach

Second stage: pool results from first stage

- multivariate meta-analysis of regression coefficients
- weighted average of predictions using *inverse-variance*
 - bootstrapping (study- and model- specific)
 - variance from auxiliary linear regression (study-specific)
- weighted summary of MCMC samples for Bayesian analyses

! Applicable second stage method varies on the basis of the first-stage model

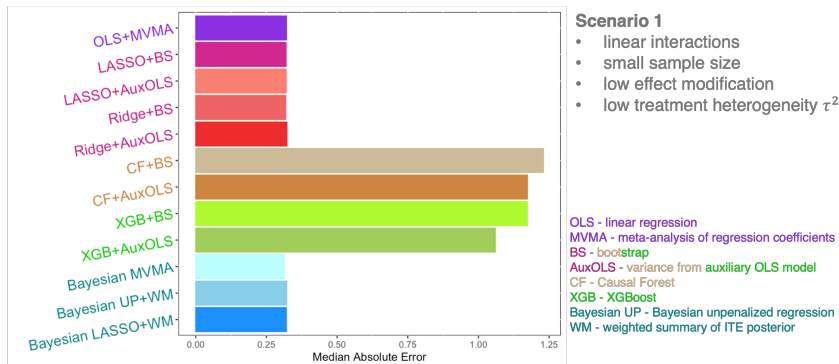
Framework pipeline: a two-stage approach



Simulation results: Models comparison

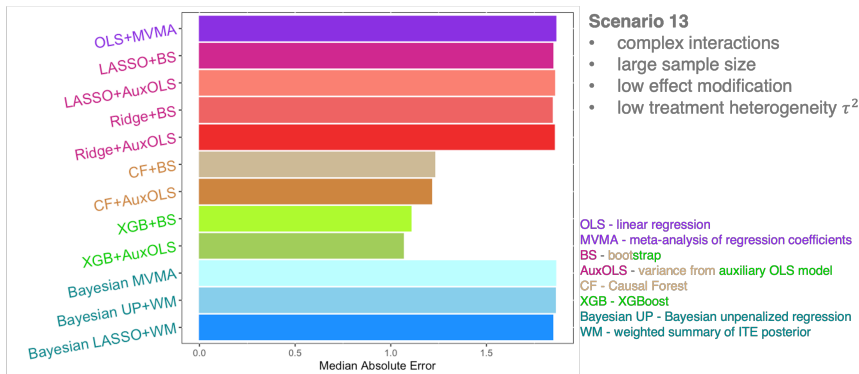
Simulation showed a **dependence** of the framework performance from the **underlying data generation mechanism**

- Given scenarios with linear interactions, regression-based models in the first stage consistently outperformed machine learning



Simulation results: Models comparison

- On the contrary, with complex interactions, machine learning approaches performed better with large samples
- In particular, weighting by OLS inverse-variance in the second stage performed better than bootstrapping



Conclusions

- This framework allows to **leverage IPD-MA** to support clinical decision making with a **flexible approach**
- Simulations showed how performance depended on the way data were generated
- For example, linear regression works well with small samples, but in some scenarios machine learning allows more accurate predictions
- However, data generation mechanisms are not known in real world applications. **Specification** of the framework and **model selection** should be based on **proper validation** (*see Orestis' talk*)

Thank you for your attention!

Questions?

Let's discuss at my poster tomorrow!
or check preprint here:



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